OBIS Code to read instantaneous parameters from NIC5 DLMS METERS

1. OBIS to read individual instantaneous parameters.

Individual voltages and currents can be read through class id 3, attribute 2 by below mentioned OBIS codes.

Parameters	CAT1	CAT2	CAT3	CAT4	CAT5b	CAT6a
Voltage Phase A	1.0.32.7.0.255	1.0.32.7.0.255	1.0.32.7.0.255	1.0.32.7.0.255	1.0.32.7.0.255	1.0.32.7.0.255
Voltage Phase B	×	×	×	1.0.52.7.0.255	1.0.52.7.0.255	1.0.52.7.0.255
Voltage Phase C	×	×	×	1.0.72.7.0.255	1.0.72.7.0.255	1.0.72.7.0.255
Line Current Phase A	1.0.31.7.0.255	×	1.0.31.7.0.255	1.0.31.7.0.255	1.0.31.7.0.255	1.0.31.7.0.255
Line Current Phase B	×	×	1.0.51.7.0.255	1.0.51.7.0.255	1.0.51.7.0.255	1.0.51.7.0.255
Line Current Phase C	×	×	×	1.0.71.7.0.255	1.0.71.7.0.255	1.0.71.7.0.255
Active Current Phase A	1.0.161.7.0.255	1.0.161.7.0.255	1.0.161.7.0.255	1.0.161.7.0.255	1.0.161.7.0.255	1.0.161.7.0.255
Active Current Phase B	×	×	1.0.161.7.0.255	1.0.163.7.0.255	1.0.163.7.0.255	1.0.163.7.0.255
Active Current Phase C	×	×	×	1.0.165.7.0.255	1.0.165.7.0.255	1.0.165.7.0.255
Line Current Summation	×	1.0.31.7.0.255	×	×	×	×
Active Current Summation	×	1.0.31.7.0.255	×	×	×	×

Line current read by above OBIS will have signed values. Amplitude will be of line current, and sign will be of active current. Positive values denote import (From grid to meter) and negative value denotes export (meter to Grid).

2. Other and more effective way of reading instantaneous parameter from meter through single OBIS code.

Alternatively, all instantaneous parameters can be read through single OBIS code by profile class.

Class ID = 7
OBIS = 1.0.94.91.0.255
Attribute = 2 for value and 3 for the OBIS code of the parameters.
The sequence of OBIS list returned in response of attribute 3 and the sequence of values returned in attribute 2 response will be identical. Line current values are unsigned values in profile data
Below table provides the details of the parameters returned in response.

OBIS Code to read instantaneous parameters from NIC5 DLMS METERS

Table 1 Parameter List of Instantaneous Profile

Parameter name	OBIS code	Remarks
Time	0.0.1.0.0.255	
Voltage-L1	1.0.32.7.0.255	
Voltage-L2	1.0.52.7.0.255	3p only
Voltage-L3	1.0.72.7.0.255	3p only
Line Current-L1	1.0.31.7.0.255	
Line Current-L2	1.0.51.7.0.255	1P2E and 3P
Line Current-L3	1.0.71.7.0.255	3p only
Active Current-L1	1.0.161.7.0.255	
Active Current-L2	1.0.163.7.0.255	1P2E and 3P
Active Current-L3	1.0.165.7.0.255	3p only
Reactive Current-L1	1.0.162.7.0.255	
Reactive Current-L2	1.0.164.7.0.255	1P2E and 3P
Reactive Current-L3	1.0.166.7.0.255	3p only
Power Factor-L1	1.0.33.7.0.255	
Power Factor-L2	1.0.53.7.0.255	3p only
Power Factor-L3	1.0.73.7.0.255	3p only
Active power Import	1.0.1.7.0.255	
Active Power Export	1.0.2.7.0.255	
Reactive Power - Q1	1.0.5.7.0.255	
Reactive Power - Q2	1.0.6.7.0.255	
Reactive Power - Q3	1.0.7.7.0.255	
Reactive Power - Q4	1.0.8.7.0.255	
Apparent power	1.0.9.7.0.255	
Average pf	1.0.13.7.0.255	
Frequency	1.0.14.7.0.255	
Phase Angle between R & Y	1.0.81.7.1.255	
Phase Angle between Y & B	1.0.81.7.12.255	3p only
Phase Angle between R & B	1.0.81.7.20.255	3p only
Cumulative Billing Count	1.0.0.1.0.255	